

IEEE WCCI/CEC 2026

Competition on “numerical optimization considering accuracy and speed”

Kangjia Qiao, Xuanxuan Ban, Peng Chen, Hongyu Lin, Kenneth V. Price,
Ponnuthurai N. Suganthan, Jing Liang, Caitong Yue, Guohua Wu

Cite this PPT presentation as:

Kangjia Qiao, Xuanxuan Ban, Peng Chen, Kenneth V. Price, Ponnuthurai N. Suganthan, Jing Liang, Caitong Yue, Guohua Wu,
“Performance comparison of CEC 2026 competition entries on numerical optimization considering accuracy and speed,”
Technical Report, Zhengzhou University, Qatar University, 2026.

Reference:

KV Price, A Kumar, PN Suganthan, “Trial-based dominance for comparing both the speed and accuracy of stochastic optimizers with standard non-parametric tests”, Swarm and Evolutionary Computation, 78, 101287, 2023.

Contents

- **Competition 1-BC-SOPs**
- Competition 2-CSOPs
- Competition 3-BC-MOPs
- Competition 4-CMOPs
- Ranking result

Competition 1-BC-SOPs

Bound constrained single objective optimization problems (BC-SOPs)

Test Problems: The 29 real-parameter numerical optimization problems with $30D$ in CEC2017 [1] are adopted as test problems.

Number of Trials/Problem: 25 independent runs.

Maximum Number of Function Evaluations: $\text{Max_FEs} = 10000 * D$, where D is the dimensionality of the optimization problems.

Population Size: You are free to have an appropriate population size to suit your algorithm while not exceeding the Max_FEs .

Sampling Points: The best EV (Error Value) every $10 * D$ evaluations will be recorded for each run. For example, the maximum number of function evaluations Max_FEs is $10000 * D$, then 1000 EVs should be saved

Target Error Values: The target error value, TGT_EV for each problem, will be determined after the competition. Hence, all algorithms should be executed until the Maximum number of Function Evaluations (Max_FEs) are consumed.

Competition 1-BC-SOPs

Bound constrained single objective optimization problems (BC-SOPs)

Presentation of Results: The results can be saved in the form of Table 1, where Min_EV is the best error value of each run at each sampling point. The value should be recorded every $10 \cdot D$ FEs. Thus, for each algorithm, 29 files should be zipped and sent to the organizers, where 29 represents the total number of test functions.

Table 1

Results saved in "PaperID_FJ_Min_EV.mat" where $J=1,2,3,\dots,29$ problems.

	Run 1	Run 2	Run 3	...	Run 25
Min_EV at Initialisation FEs					
Min_EV at $10 \cdot D$ FEs					
Min_EV at $20 \cdot D$ FEs					
...					
...					
Min_EV at Max_FEs					

Competition 1-BC-SOPs

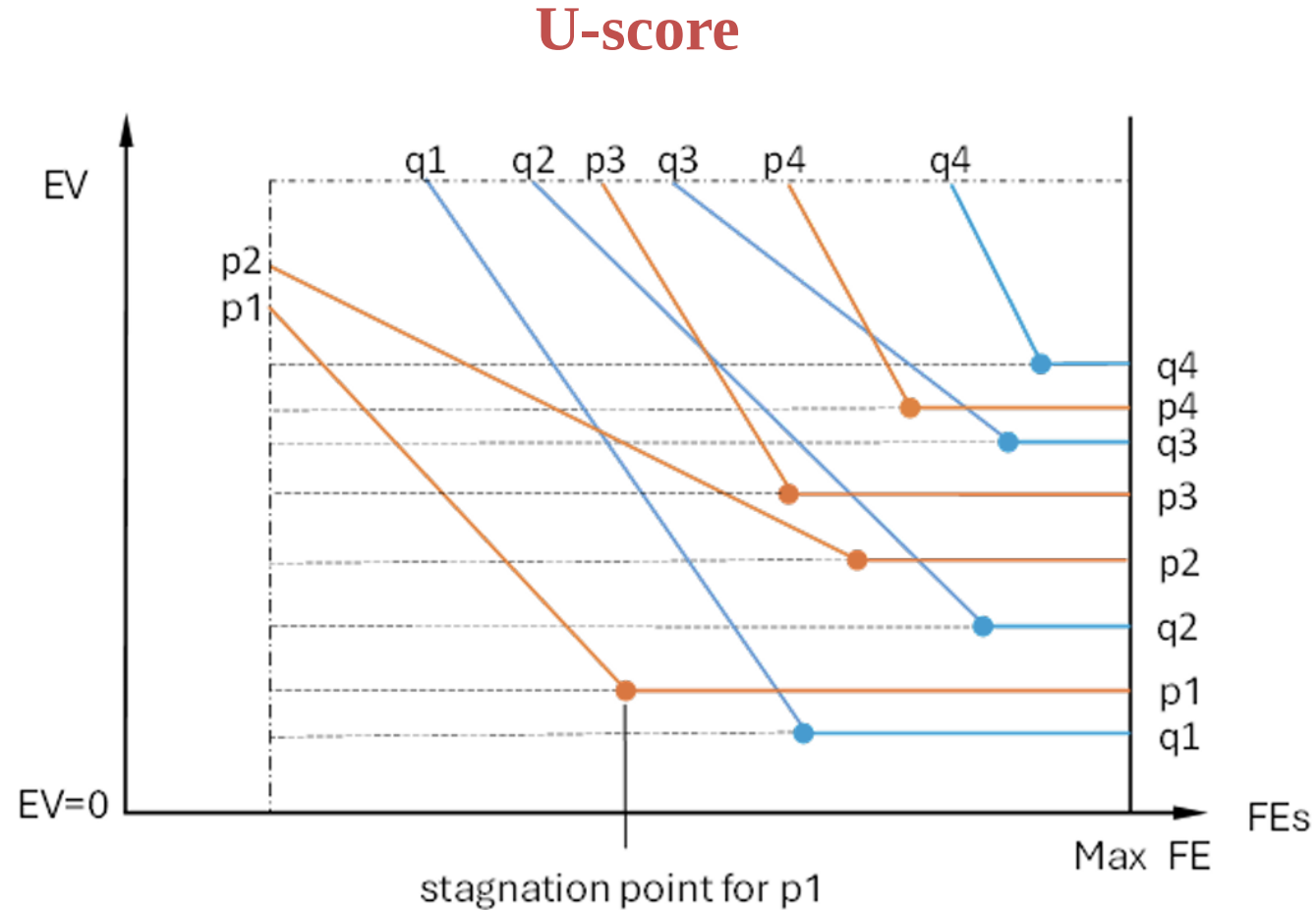


Fig. 1: A convergence plot showing 8 trials from two algorithms and their stagnation points.

Contents

- Competition 1-BC-SOPs
- **Competition 2-CSOPs**
- Competition 3-BC-MOPs
- Competition 4-CMOPs
- Ranking result

Competition 2-CSOPs

Constrained single objective optimization problems (CSOPs)

Test Problems: The 28 constrained real-parameter optimization problems with 30D in CEC2017 [2] are adopted as test problems.

Number of Trials/Problem: 25 independent runs.

Maximum Number of Function Evaluations: $\text{Max_FEs} = 20000 * D$, where D is the dimensionality of the optimization problems.

Population Size: You are free to have an appropriate population size to suit your algorithm while not exceeding the Max_FEs .

Sampling Points: Record f_{min} values and LCV every $10 * D$ evaluations. For example, if the maximum number of function evaluations Max_FEs is $20000 * D$, then 2000 f_{min} values are recorded for trials with one or more feasible solutions. When the whole population is infeasible, the lowest LCV value of the population should be saved at the respective sampling points.

Target Error Values: The target error value will be determined after the competition. Hence, all algorithms should be executed until Maximum number of Function Evaluations (Max_FEs) are consumed.

Competition 2-CSOPs

Constrained single objective optimization problems (CSOPs)

Presentation of Results: Save your results as shown in Table 2, in which the first entry is for the evaluation of the initial population. The cumulative FEs at each sampling point should be saved in the first column. Meanwhile, the corresponding f_{min} and LCV results should be saved in the second and third columns, respectively. So, for a function, one run requires one file in mat format. Please note that if no feasible solution exists at one sampling point, the f_{min} result should be expressed by "NaN".

$$LCV = \min: CV(P_i), i = 1, \dots, NP$$

Table 2

Results saved in "PaperID_CPJ.mat" where J=1,2,...,28 problems

FEs	Run1		Run2		...	Run25	
	f_{min}	LCV	f_{min}	LCV		f_{min}	LCV
at Initialisation FEs							
Sampling Point 1, FEs=1*10D							
Sampling Point 2, FEs=2*10D							
...							
Last Sampling Point, Max_FEs							

Competition 2-CSOPs

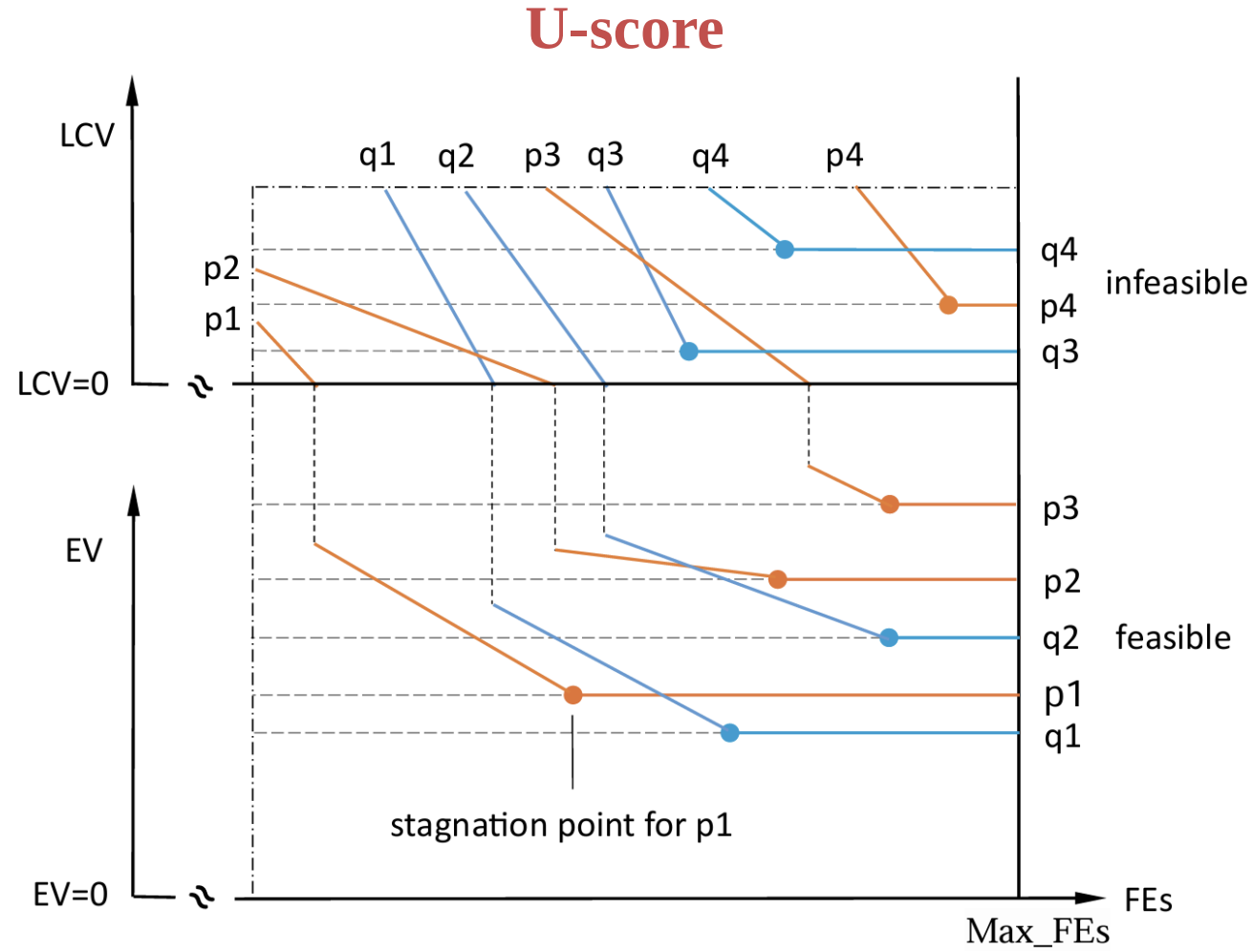


Fig. 1: A convergence plot showing 8 trials from two algorithms and their stagnation points. In this example, all trials are initially infeasible, but only 5 become feasible before reaching Max_FEs.

Contents

- Competition 1-BC-SOPs
- Competition 2-CSOPs
- **Competition 3-BC-MOPs**
- Competition 4-CMOPs
- Ranking result

Competition 3-BC-MOPs

Bound constrained multi-objective optimization problems (BC-MOPs)

Test Problems: We adopt the benchmark of [3] including 10 multi-objective problems to rank the optimizers of MOPs without constraints.

Number of Trials/Problem: 30 independent runs.

Maximum Number of Function Evaluations: 100000 for each function.

Population Size: 100.

Sampling Points: The *IGD* values will be recorded once every 200 function evaluations. For example, if the maximum number of evaluations *Max_FEs* is 100000, then 500 *IGD* values are saved.

Target Error Values: The target *IGD* value will be determined after the competition. Hence, all algorithms should be executed until Maximum number of Function Evaluations (*Max_FEs*) are consumed.

Competition 3-BC-MOPs

Bound constrained multi-objective optimization problems (BC-MOPs)

Presentation of Results: To compare and evaluate the algorithms participating in the competition, it is necessary that the authors email the results as shown in Table 4 to the organizers after submitting the final version of the accepted paper.

Table 4

Results saved in "PaperID RCMJ IGD.txt" where $J=1,2,\dots,10$ problems

	Run 1	Run 2	Run 3	...	Run 30
IGD at Initialisation FEs					
IGD at Sampling Point 1					
IGD at Sampling Point 2					
...					
...					
IGD at Sampling Point 500, 100K FEs					

Competition 3-BC-MOPs

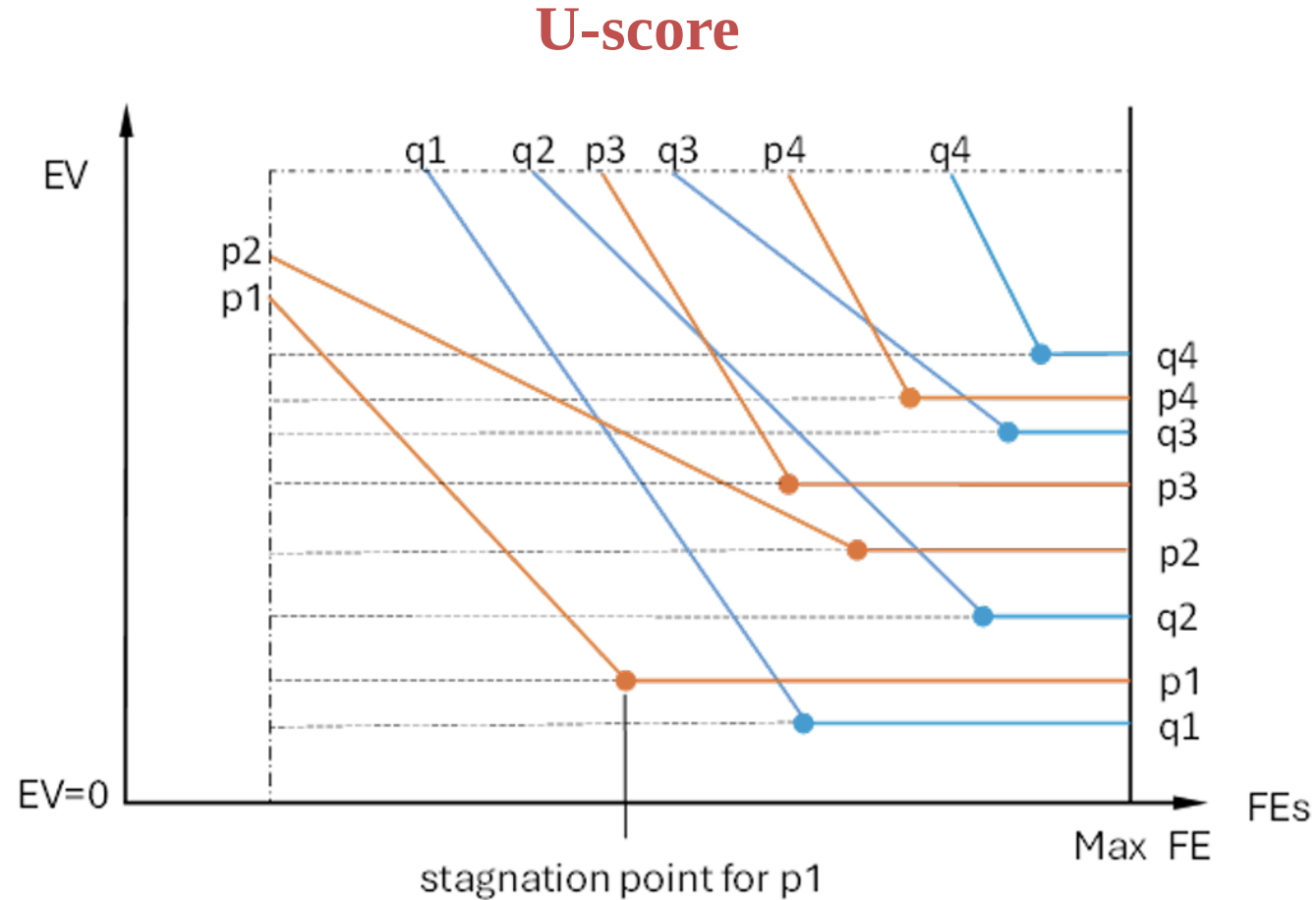


Fig. 1: A convergence plot showing 8 trials from two algorithms and their stagnation points.

Contents

- Competition 1-BC-SOPs
- Competition 2-CSOPs
- Competition 3-BC-MOPs
- **Competition 4-CMOPs**
- Ranking result

Competition 4-CMOPs

Constrained multi-objective optimization problems (CMOPs)

Test Problems: The latest constrained multiobjective optimization problems with scalable decision space constraints (SDC problems) [4] are adopted as test problems. SDC benchmark contains 15 problems.

Number of Trials/Problem: 30 independent runs.

Maximum Number of Function Evaluations: 200000 for each function.

Population Size: 100. **Dimension:** 30 for each SDC function.

Sampling Points: The IGD values will be recorded once every 200 function evaluations. For example, if the maximum number of evaluations Max_FEs is 200000, then 1000 IGD values are saved.

Target Error Values: The target IGD value will be determined after the competition. Hence, all algorithms should be executed until the Maximum number of Function Evaluations (Max_FEs) are consumed. Please note that the minimal IGD value is unknown for multiobjective optimization problems. So, the mean or median IGD value of all trials from all algorithms participating in the competition will be set as the target IGD value

Competition 4-CMOPs

Constrained multi-objective optimization problems (CMOPs)

Presentation of Results: To compare and evaluate the algorithms participating in the competition, it is necessary that the authors email the results in the format as shown in Table 5 to the organizers, after submitting the final version of the accepted papers.

$$MCV = \frac{\sum_{i=1}^{PF} CV(P_i)}{PF}$$

Table 5

Results saved in "PaperID_SDCJ.mat" where J=1,2,...,15 problems.

	Run1		Run2		...	Run 30	
	<i>IGD</i>	<i>MCV</i>	<i>IGD</i>	<i>MCV</i>		<i>IGD</i>	<i>MCV</i>
at initialization FEs							
Sampling point 1							
Sampling point 2							
...							
...							
Sampling point 1000							

Competition 4-CMOPs

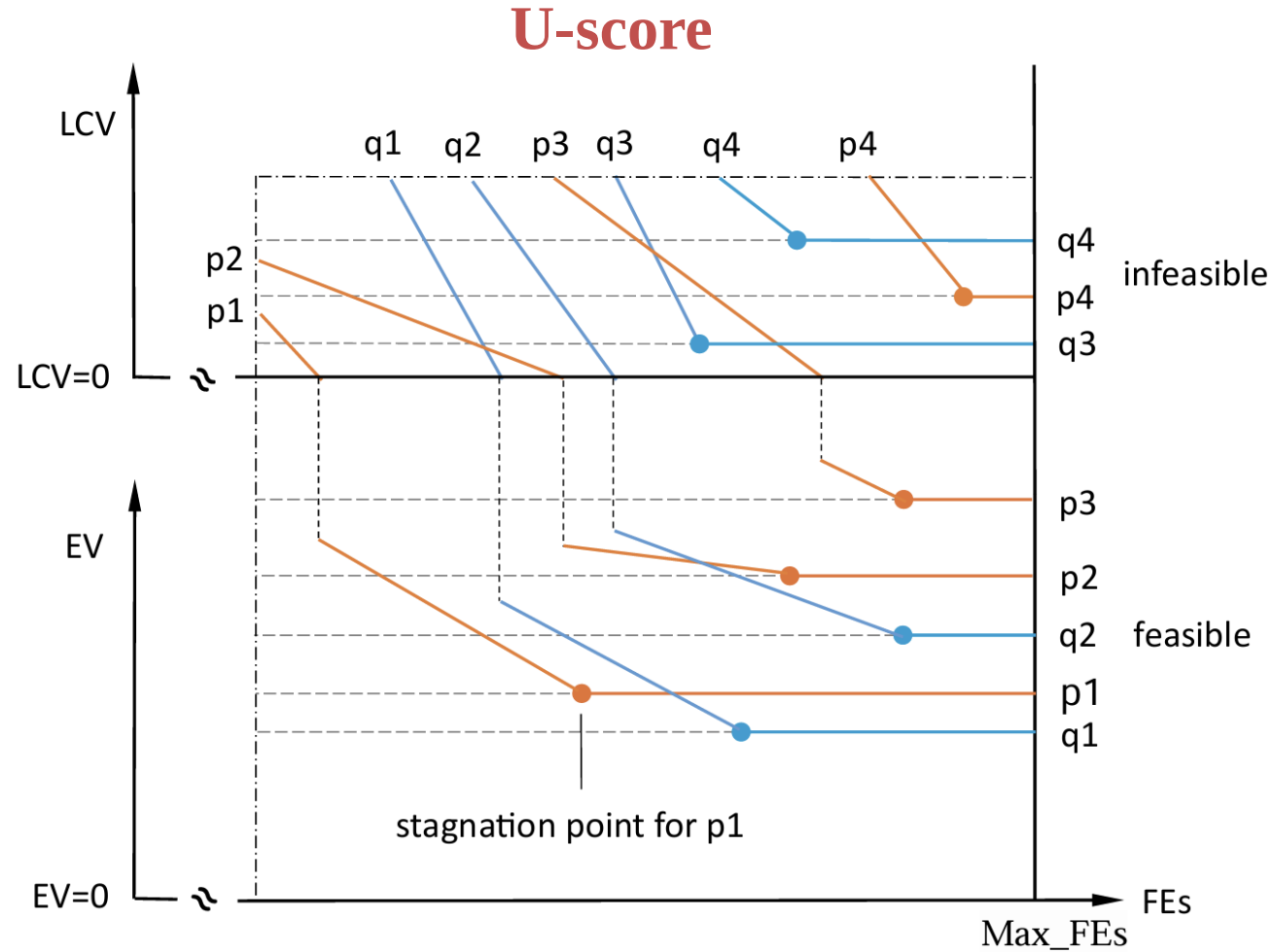


Fig. 1: A convergence plot showing 8 trials from two algorithms and their stagnation points. In this example, all trials are initially infeasible, but only 5 become feasible before reaching Max_FEs.

Contents

- Introduction
- Test problems suite
- Indicators and rules
- Participators
- **Ranking result**

Competition 1-BC-SOPs

Algorithm	Ranking	Authors	Paper title/link
RDE26	 1	Sichen Tao, Hanyu Hu, Ruihan Zhao, Qingke Zhang, Yifei Yang, Jian Wang, Masatoshi Kawai, Hiroyuki Takizawa	https://github.com/SichenTao/RDE26-SOP
DE-2LS	 2	J Senthilnath, Anupam Trivedi, Dikshit Chauhan, Chen Hao, Harikumar Kandath	https://github.com/pingu-73/RDEx-CASK/tree/main/Report
RDEx	 3	Sichen Tao, Yifei Yang, Ruihan Zhao et al.	https://github.com/SichenTao/IEEE-CEC-2025-Competition-RDEx/tree/main
L-SRTDE	4	Vladimir Stanovov Eugene Semenkin	Success Rate-based Adaptive Differential Evolution L-SRTDE for CEC 2024 Competition, In Proc. CEC 2024
rcmaes	5	Khoirul Faiq Muzakka	https://github.com/khoirulmuzakka/Minion/blob/main/examples/main_cec_multi.cpp
R-CMA-ES	6	Adam Stelmaszczyk	Refined CMA-ES for CEC 2026 Bound-Constrained Single-Objective Optimization, In Proc. CEC 2026.
mLSHADE_LR	7	Dikshit Chauhan, Anupam Trivedi, Shivani	https://arxiv.org/abs/2409.15994
BlockEA	8	Xuhong Qi	https://github.com/QiXuhong520/Algorithm-report
jSOa	9	Petr Bujok	https://github.com/PetBuj/jSOa/blob/main/jSOaGitHub.pdf
IEACOP	10	Andrea Tangherloni Vasco Coelho Francesca M. Buffa Paolo Cazzaniga	A modified EACOP implementation for Real-Parameter Single Objective Optimization Problems, In Proc. CEC 2024
ADSDE	11	Jiarui Yang, Mingjing Wang	

Competition 1-BC-SOPs

Prob\Algo	RDE26	DE-2LS	RDEx	L-SRTDE	rcmaes	R-CMA-ES	mLSHADE_L R	BlockEA	jSOa	IEACOP	ADSDE
F1	9756.5 /2	8759.5 /3	8161.5 /4	7499 /5	4550 /10	11325 /1	6203.5 /6	2468 /11	6124 /7	5403 /8	5100 /9
F2	11379.5 /1	10109.5 /2	9575.5 /3	8887.5 /4	6496.5 /7	7475 /5	5726.5 /8	2460 /10	7037 /6	4999 /9	1204 /11
F3	7082 /6	7152.5 /5	7757.5 /3	5998 /9	5857 /10	7475 /4	8621 /2	6437.5 /7	6053 /8	11704 /1	1212.5 /11
F4	9560 /2	9537.5 /3	8920.5 /5	9372.5 /4	12083.5 /1	7475 /6	5851 /7	4127.5 /9	4728 /8	3094.5 /10	600 /11
F5	10411.5 /1	9570.5 /2	7998.5 /5	9204.5 /4	6981 /7	5345.5 /8	9298 /3	5223 /9	7941.5 /6	2776 /10	600 /11
F6	9090 /3	7955 /4	7902 /5	6501 /7	11021 /2	6850 /6	4659 /8	13100 /1	4003 /9	3669 /10	600 /11
F7	10089.5 /2	9743 /3	8820 /5	8893 /4	11878 /1	7475 /6	5767 /7	3809 /9	4567 /8	3708.5 /10	600 /11
F8	9646 /2	8699 /4	8155 /5	10614 /1	5009 /8	7475 /6	9289.5 /3	2936 /10	7270.5 /7	3417 /9	2839 /11
F9	8231 /4	7872 /5	7363 /7	7551.5 /6	10982 /2	8262 /3	3804 /9	12416 /1	3545 /10	4651.5 /8	672 /11
F10	9640.5 /3	9515.5 /4	9948.5 /2	10822.5 /1	7097.5 /6	8100 /5	5643 /9	1376 /10	6305.5 /7	5827 /8	1074 /11
F11	10388.5 /1	10381.5 /2	10338 /3	9754.5 /4	6587.5 /7	7942 /5	2846 /10	7401 /6	5528 /8	3583 /9	600 /11
F12	10016 /2	9760.5 /3	10287 /1	9290 /5	9598 /4	7930 /6	5980 /7	2279.5 /10	5519 /8	3747.5 /9	942.5 /11
F13	10790.5 /1	10480 /2	9834.5 /3	8913.5 /4	8661 /5	7475 /6	5178 /8	2550 /10	7162.5 /7	3155 /9	1150 /11
F14	10796.5 /2	10914 /1	10507.5 /3	8334.5 /5	8928.5 /4	7545 /6	4969.5 /8	2343.5 /10	6261 /7	3661 /9	1089 /11
F15	10300.5 /1	10183 /3	9893.5 /4	10262.5 /2	8129.5 /5	8069 /6	5947.5 /7	4140.5 /9	4703.5 /8	3120.5 /10	600 /11

Competition 1-BC-SOPs

Prob\Algo	RDE26	DE-2LS	RDEx	L-SRTDE	rcmaes	R-CMA-ES	mLSHADE_L R	BlockEA	jSOa	IEACOP	ADSDE
F16	10930 /2	9858.5 /3	9085.5 /4	8052.5 /5	11126.5 /1	7660 /6	5142.5 /7	4089 /9	5112.5 /8	3693 /10	600 /11
F17	10003 /3	9773 /4	10226 /1	7754 /5	10214 /2	7564 /6	4570.5 /9	4623.5 /8	7554 /7	2468 /10	600 /11
F18	11025 /1	9876 /4	10379.5 /3	10972 /2	7181.5 /6	7535 /5	4567 /8	3028 /10	6267.5 /7	3456.5 /9	1062 /11
F19	10917 /1	10116.5 /2	9952 /3	3725 /10	8630.5 /4	7909.5 /5	5612.5 /8	5984 /7	6207.5 /6	4413 /9	1882.5 /11
F20	8897.5 /3	8074.5 /6	8253 /5	8375.5 /4	10438 /2	7475 /7	4709 /8	11460.5 /1	3518.5 /10	3548.5 /9	600 /11
F21	9806 /2	8819 /3	7965 /5	7275 /7	5941 /9	7555 /6	9919 /1	2300 /10	6350 /8	8725 /4	695 /11
F22	10218.5 /1	9974 /2	9239.5 /3	8623.5 /5	9194.5 /4	7850 /6	5228 /8	5760.5 /7	4813.5 /9	3510.5 /10	937.5 /11
F23	9549 /2	8221 /6	8254 /5	8565 /4	8747 /3	7475 /7	5022 /8	12320.5 /1	3511 /9	3085.5 /10	600 /11
F24	7596.5 /4	11776.5 /1	11635.5 /2	6987.5 /7	7768.5 /3	7475 /5	7132 /6	2475 /10	6302 /8	5601.5 /9	600 /11
F25	8608.5 /3	8558 /4	9014.5 /2	7922 /7	8223 /6	8475 /5	3893 /9	11879 /1	3663 /10	4514 /8	600 /11
F26	9702 /4	11194 /2	11294 /1	9793 /3	6218 /6	7475 /5	4418.5 /9	5370 /8	5681.5 /7	3604 /10	600 /11
F27	9748 /1	9708.5 /2	9136.5 /3	8952 /4	5322.5 /9	8159 /6	6843 /7	1931 /10	6784 /8	8165.5 /5	600 /11
F28	8051 /6	8905 /2	8775 /4	12473 /1	8402.5 /5	7582 /7	4715.5 /8	8884.5 /3	4339.5 /9	2622 /10	600 /11
F29	5187.5 /9	9825.5 /3	9503.5 /4	12203 /1	6056.5 /6	8130.5 /5	5188.5 /8	10794.5 /2	5784 /7	2076.5 /10	600 /11
sum/RS	277418 /75	275313 /90	268176 /103	253571.5 /130	237324 /145	224538.5 /160	166744.5 /206	163967.5 /209	162637 /227	126000 /252	29460 /317

Competition 2-CSOPs

Algorithm	Ranking	Authors	Paper title/link
RDE26	 1	Sichen Tao, Hanyu Hu, Ruihan Zhao, Qingke Zhang, Yifei Yang, Jian Wang, Masatoshi Kawai, Hiroyuki Takizawa	Reconstructed Differential Evolution for the IEEE WCCI/CEC 2026 competition on Variable-Constrained Single-Objective Optimization Problems https://github.com/SichenTao/RDE26-CSOP
RDEx	 2	Sichen Tao, Yifei Yang, Ruihan Zhao et al.	https://github.com/SichenTao/IEEE-CEC-2025-Competition-RDEx
RDEx_CASK	 3	Dikshant, Dikshit Chauhan, Chen Hao, Anupam Trivedi, K. Harikumar, J. Senthilnath	RDEx-CASK: Cauchy Mutation, Archive, and Stagnation Kick for RDEx-CSOP. https://github.com/pingu-73/RDEx-CASK
DE-2LS	4	J Senthilnath, Anupam Trivedi, Dikshit Chauhan, Chen Hao, Harikumar Kandath	https://github.com/pingu-73/RDEx-CASK/tree/main/Report
AEBDE	5	Bingchao Shi, Chennuo Jin, Yifei Yang, Sichen Tao	Adaptive Elite Branch Differential Evolution for Constrained Single-Objective Optimization
UDE-III	6	Anupam Trivedi Dikshit Chauhan	https://arxiv.org/abs/2410.03992
UDE_IV	7	Anupam Trivedi	UDE-IV: An Enhanced Unified Differential Evolution Algorithm for CEC 2025 Constrained Optimization Problems
CL-SRDE	8	Vladimir Stanovov Eugene Semenkin	Differential Evolution with Success Rate-based adaptation CL-SRDE for Constrained Optimization, In Proc. CEC 2024.

Competition 2-CSOPs

Prob\Algo	RDE26	RDE _x	RDE _x _CASK	DE-2LS	AEBDE	UDEIII	UDE_IV	CL_SRDE
C01	7162.5/1	5159.5/4	6537.5/2	5034.5/5	5660.5/3	4045.5/6	3313/7	2887/8
C02	7162.5/1	5104.5/4	6537.5/2	5094/5	5663/3	4038/6	3386.5/7	2814/8
C03	5978.5/3	4483.5/5	3772/6	3658/7	4612/4	7809/2	8653/1	834/8
C04	7887.5/1	3946.5/6	4293/4	3690.5/7	3963.5/5	7742/2	7002/3	1275/8
C05	7162.5/1	5009.5/4	6537.5/2	5133.5/3	4779/6	4978/5	3412.5/7	2787.5/8
C06	6499/3	6742.5/1	3314.5/7	6670/2	2844/8	5157.5/4	4199/6	4373.5/5
C07	5169.5/4	8020.5/1	5259/3	7590/2	1850/8	3506/6	3492/7	4913/5
C08	7406.5/2	6071.5/3	8100/1	5823.5/4	3723.5/6	2435.5/7	1264.5/8	4975/5
C09	6501/4	7034.5/2	6704.5/3	7047.5/1	3530.5/6	2464.5/7	1643.5/8	4874/5
C10	7399.5/1	6049/4	7394/2	6152.5/3	4130/6	2475/7	1225/8	4975/5
C11	8026.5/1	5521/5	6033.5/4	6787/2	3051/6	2536.5/7	1059/8	6785.5/3
C12	6566.5/3	4074/5	2677/8	3858/6	4231/4	7758/1	7635.5/2	3000/7
C13	7225/1	5074/4	6097/2	5022.5/5	5607/3	4183/6	3517/7	3074.5/8
C14	6267/1	5030.5/5	6142.5/2	4950.5/6	4802.5/7	5819.5/3	5171/4	1616.5/8

Competition 2-CSOPs

Prob\Algo	RDE26	RDE _x	RDE _x _CASK	DE-2LS	AEBDE	UDEIII	UDE_IV	CL_SRDE
C15	8544/1	8513.5/2	1692/7	1425.5/8	4758/4	4072/5	4018/6	6777/3
C16	4918/3	4620/4	2364.5/7	4373.5/5	3997.5/6	8689/2	8740/1	2097.5/8
C17	6587.5/1	4433.5/6	5337.5/4	4366.5/7	5962.5/2	4461/5	5354/3	3297.5/8
C18	7686.5/1	6174.5/3	6209.5/2	6098/4	5997.5/5	743.5/8	2291/7	4599.5/6
C19	5782/2	4910.5/5	5516/3	5112/4	6225/1	4496.5/6	3985.5/7	3772.5/8
C20	8364/1	6956/2	4695/5	6424/3	6307/4	2484.5/7	2764.5/6	1805/8
C21	5853/4	4130.5/5	6494/3	3776/6	3139/8	6514/2	6589/1	3304.5/7
C22	7675/1	5969/4	5226/5	6041.5/3	6140.5/2	2801/7	1247/8	4700/6
C23	6818/1	5339/3	6457.5/2	5054.5/4	5018.5/5	2913.5/8	4361.5/6	3837.5/7
C24	8399.5/1	8391.5/2	2087.5/7	1702/8	4725/4	3556/6	3657/5	7281.5/3
C25	5347.5/4	5444/3	1506.5/8	5037/5	4962.5/6	7923/1	7011.5/2	2568/7
C26	7050/1	4851.5/5	5693.5/3	4921.5/4	6425/2	3541.5/7	3441.5/8	3875.5/6
C27	8377.5/1	5468/5	6639.5/2	5537.5/3	5102.5/6	896/8	2293.5/7	5485.5/4
C28	4937.5/6	6448.5/3	5870.5/5	6764.5/2	6388.5/4	1497/7	953/8	6940.5/1
sum/RS	192754/55	158971/105	145189/111	143146/124	133596/134	119536/148	111680/158	109526/173

Competition 3-BC-MOPs

Algorithm	Ranking	Authors	Paper title/link
RDE26	 1	Sichen Tao Hanyu Hu Ruihan Zhao Qingke Zhang Yifei Yang Jian Wang Masatoshi Kawai Hiroyuki Takizawa	Reconstructed Differential Evolution for the IEEE WCCI/CEC 2026 competition on Bound-Constrained Multi-Objective Optimization Problems https://github.com/SichenTao/RDE26-MOP
RDE-SOR	 2	Zhengzhang Zhao Hanyu Hu Ziwei Bian Xiaodong Huang Sichen Tao Jian Wang	A State-Driven Reconstructed Differential Evolution Variant for the IEEE WCCI/CEC 2026 competition on Bound-Constrained Multi-Objective Optimization Problems https://github.com/SichenTao/RDE-SOR
RDE_x	 3	Sichen Tao, Yifei Yang, Ruihan Zhao et al.	Efficient Reconstructed Differential Evolution for IEEE CEC 2025 Numerical Optimization Competitions https://github.com/SichenTao/IEEE-CEC-2025-Competition-RDEx

Competition 3-BC-MOPs

Algorithm	Ranking	Authors	Paper title/link
TFBCEIBEA	4	Peng Chen Jing Liang Kangjia Qia Ponnuthurai Nagaratnam Suganthan Xuanxuan Ban	A Two-stage Evolutionary Framework For Multi-objective Optimization, In proc. CEC 2024.
TEMOFNSGA3	5	Peng Chen	A Two-stage Evolutionary Framework For Multi-objective Optimization
TGFMMOEA	6	Peng Chen	A Two-stage Evolutionary Framework For Multi-objective Optimization

Competition 3-BC-MOPs

Prob\Algo	RDE26	RDE-SOR	RDE _x	TFBCEIBEA	TEMOFNSGA3	TEMOFGFMMOEA
MaOP1	9111.5/1	7593.5/2	7505/3	3736.5/4	2309/5	1964.5/6
MaOP2	9430/1	7056/3	7094/2	3058/4	3049/5	2533/6
MaOP3	9013.5/1	7310/3	7886.5/2	2475.5/5	3225/4	2309.5/6
MaOP4	7548.5/3	8328/2	8333.5/1	2095.5/6	3280/4	2634.5/5
MaOP5	9237.5/1	7571.5/2	7394.5/3	3121/4	2890.5/5	2005/6
MaOP6	9283/1	7249/3	7633/2	3916.5/4	2588/5	1550.5/6
MaOP7	8991/1	7767/2	7452/3	2836.5/5	2331.5/6	2842/4
MaOP8	9569.5/1	7396/2	7244.5/3	3097/4	2042.5/6	2870.5/5
MaOP9	7168.5/3	8571.5/1	8470/2	2585/5	2331.5/6	3093.5/4
MaOP10	7908.5/3	8132/1	8113.5/2	2858.5/5	2290.5/6	2917/4
sum/RS	87261.5/16	76974.5/21	77126.5/23	29780/46	26337.5/52	24720/52

Competition 4-CMOPs

Algorithm	Ranking	Authors	Paper title/link
RDE26	 1	Sichen Tao, Hanyu Hu, Ruihan Zhao, Qingke Zhang, Yifei Yang, Jian Wang, Masatoshi Kawai, Hiroyuki Takizawa	Reconstructed Differential Evolution for IEEE WCCI/CEC 2026 on Variable-Constrained Multi-Objective Optimization Problems. https://github.com/SichenTao/RDE26-CMOP
CLSHRDE	 2	Chennuo Jin	Closed-Loop Success-History Driven Reconstructed Differential Evolution for Constrained Multi-Objective Optimization. https://github.com/EureKaGin/CEC26_CLSHRDE
RDE _x	 3	Sichen Tao, Yifei Yang, Ruihan Zhao et al.	https://github.com/SichenTao/IEEE-CEC-2025-Competition-RDEx/
PRCEA	4	Xiaoyu Zhong	https://github.com/rotarygirl-zxy/PRCEA
DESDE	5	Kangjia Qiao	https://github.com/xxuanban/DESDE/blob/main/DESDE.pdf
CCEMT	6	Xiaoyu Zhong	https://github.com/wcq1536113693/zxyCCEMT
IMTCMO	7	Dezheng Zhang	https://github.com/DezhengZ/Algorithm-Description-IMTCMO
CCPTEA	8	Lianhe Duan	https://github.com/LianheDuan/Algorithm-Description.git
ZCZU	9	Changzhe Zheng	https://github.com/zcz2000jjhh-hub/ZCZU_CEC2026.git
MTCMMO	10	Wenhao Wu	https://github.com/zaishuiyifang1507/A-noverl-genetic-algorithm-forCEC2024 ₂₈

Competition 4-CMOPs

Prob\Algo	CCEMT	CCPTEA	CLSHRDE	DESDE	IMTCMO	MTCMMO	PRCEA	rde26-cmop	rdex-cmop	ZCZU
SDC1	12788.5/2	9554/5	9397.5/6	12362/3	10160.5/4	1325/10	12894/1	9185.5/7	7610.5/8	4422.5/9
SDC2	9200/5	8169/8	12841/2	11296.5/3	8252.5/7	870/10	8829.5/6	14124/1	11226/4	4891.5/9
SDC3	9102/5	7204/7	15473/1	12193/4	8020.5/6	6878/8	2848.5/9	13656/2	12976/3	1349/10
SDC4	7740.5/8	9309/7	11242/3	10385/5	9508/6	1323/10	5133.5/9	12714/1	11791.5/2	10553.5/4
SDC5	8304.5/6	7883.5/7	12349/3	11825/5	7198/8	3637/9	11925/4	12716.5/1	12520/2	1341.5/10
SDC6	8229/7	7215.5/9	12476/2	8827.5/5	7613.5/8	3172/10	8866/4	13440.5/1	11580/3	8280/6
SDC7	10087/5	8384/7	12916.5/2	7067/8	9983/6	870/10	13383/1	12209/3	11261.5/4	3539/9
SDC8	12073/2	11957.5/4	6251.5/9	11217.5/5	11968/3	870/10	9087/6	7470.5/7	6684.5/8	12120.5/1

Competition 4-CMOPs

Prob\Algo	CCEMT	CCPTEA	CLSHRDE	DESDE	IMTCMO	MTCMMO	PRCEA	rde26-cmop	rdex-cmop	ZCZU
SDC9	5460/8	6363.5/7	14614/2	13352.5/4	6663/5	2986.5/10	6473.5/6	15402/1	14047/3	4338/9
SDC10	9036.5/5	8203.5/7	8991.5/6	10189.5/4	7612.5/8	5565.5/9	11629.5/2	12571.5/1	11004.5/3	4895.5/10
SDC11	7918/7	7768.5/8	13258/2	11250/4	7921.5/6	878/10	8940/5	14134.5/1	12754.5/3	4877/9
SDC12	12292.5/2	10513/5	7100/8	6921.5/9	11641/3	870/10	13420/1	8429.5/6	7560/7	10952.5/4
SDC13	5152.5/8	6097/6	14980.5/1	12834/4	5823.5/7	4803.5/9	6408/5	14799.5/2	14050/3	4751.5/10
SDC14	6318/7	6333/6	13567.5/3	11386.5/4	5688/9	6178/8	7430/5	13842/2	13985/1	4972/10
SDC15	6937/7	4794.5/8	16464/1	7408.5/6	4507/9	1450/10	11252/4	14145.5/2	13235/3	9506.5/5
sum/ RS	130639/84	119750/101	181922/51	158516/73	122560/95	41676.5/143	138520/68	188840/38	172286/57	90790.5/115

Verification

Ranking related codes and data are made available online

Codes of top performing algorithms will be released online.

We will verify over the next few weeks.

Thanks for your attention!

Questions?